Catching a Quantum Spy

BB84 quantum key distribution with Qiskit



Results: Qiskit Aer simulator

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Can Alice and Bob catch a spy?

1

The problem

Alice wants to share a secret key with Bob, but Eve may be listening on the line.

2

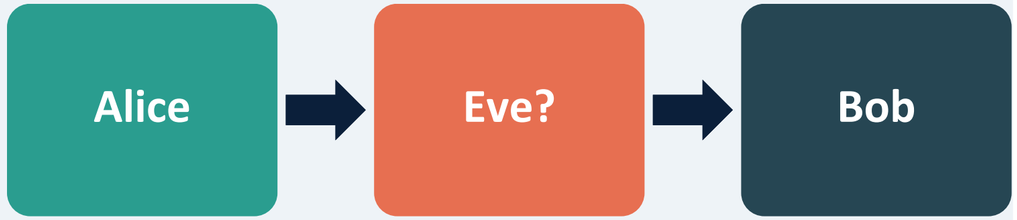
The quantum idea

Measuring a qubit in the wrong basis disturbs it, so spying leaves a trace.

3

Why I chose it

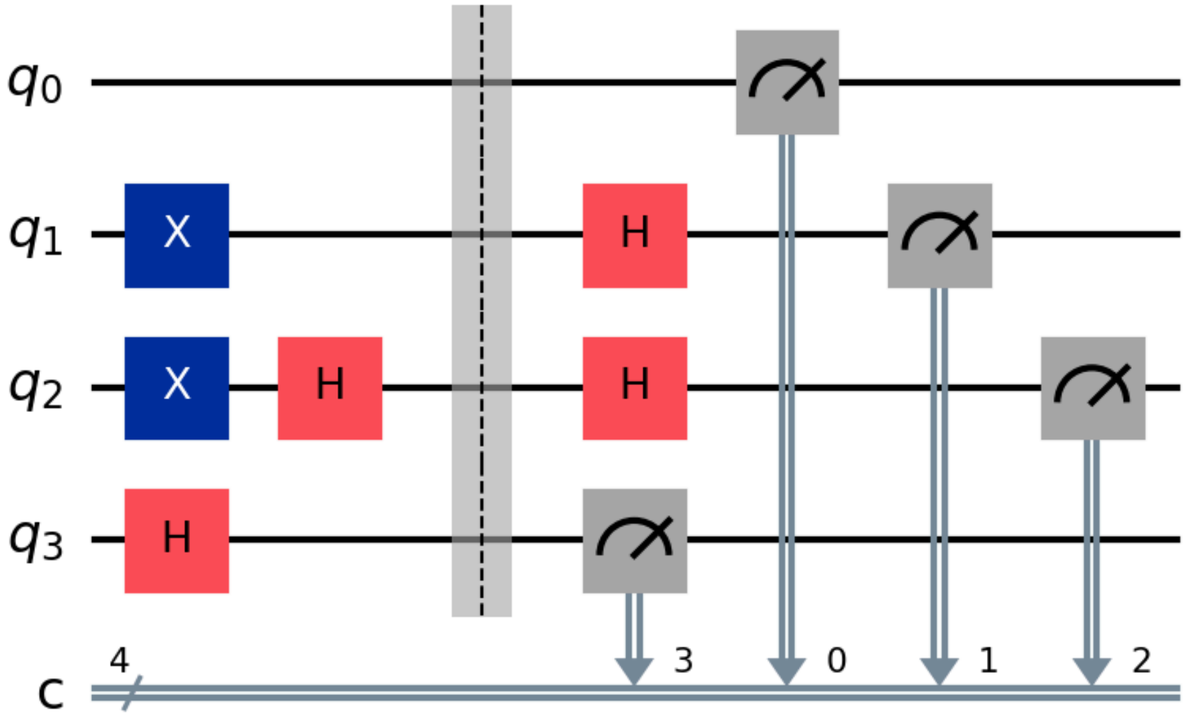
A tiny circuit (X, H, measure) gives a clear, measurable result.



Qubits travel left to right. Alice and Bob then compare bases (never bits) and keep matching ones.

No Eve: keys match.
Eve: errors appear.

How the circuit works



4-qubit example (simulator). q0, q2: same basis, kept. q1, q3: different basis, discarded.

1. **X gate**
   encodes the bit value 1

2. **H gate**
   encodes the X basis choice

3. **Measure**
   Bob reads 0 or 1 (H first if he uses the X basis)

4. **Sifting**
   keep positions where the bases match, about 50%

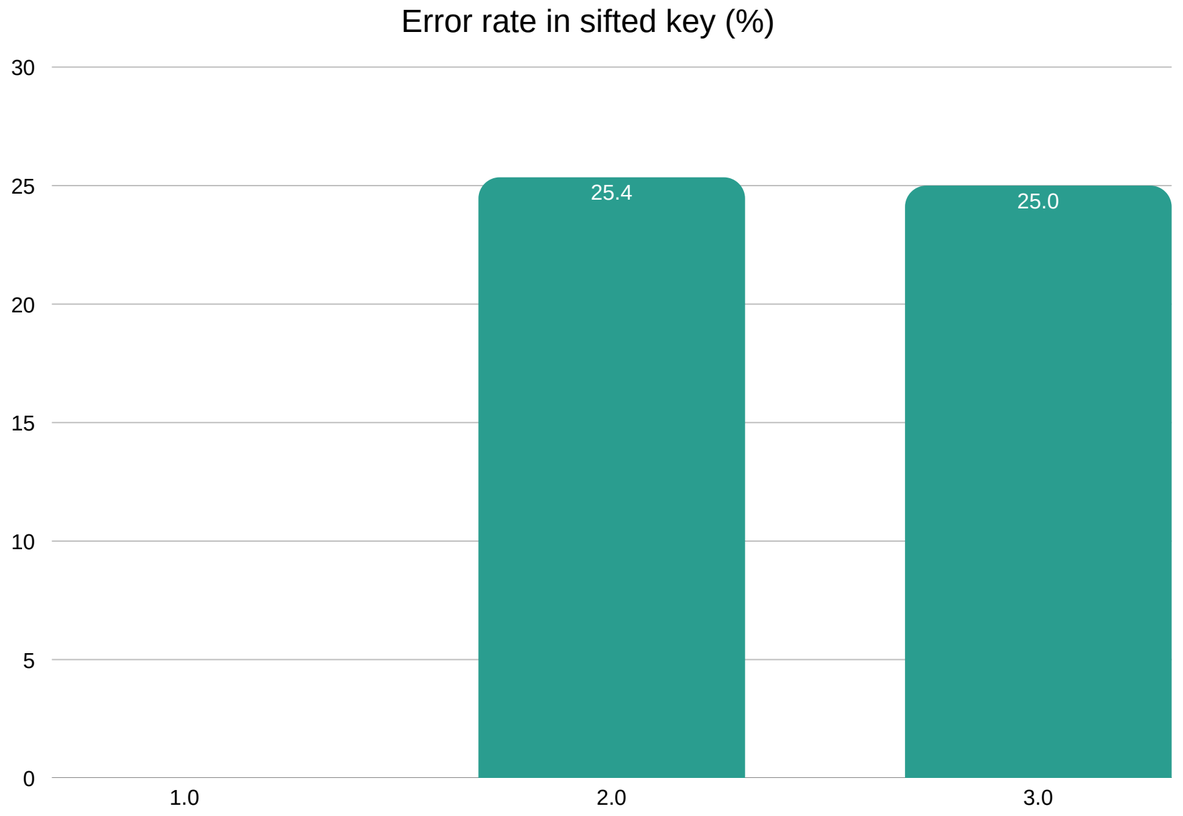
What I observed

0.00%

errors with no Eve: sifted keys are identical

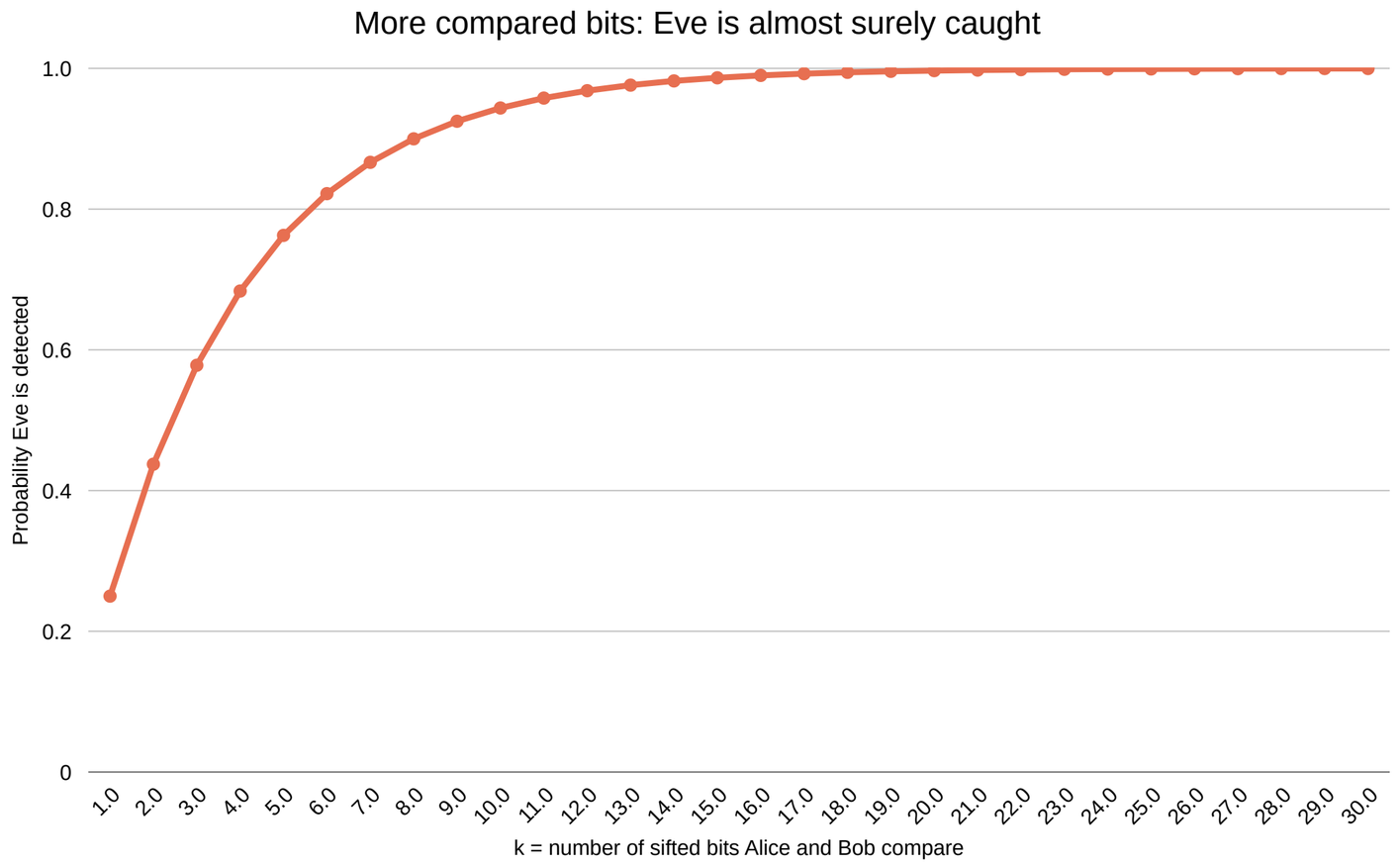
25.35%

errors when Eve intercepts every qubit (theory: 25%)



Aer simulator | 1000 qubits | seed 42 | 501 sifted bits

Why 25%, and how fast Eve is caught



Simulator dots match the theory curve 1 - (3/4)^k

1/2 x 1/2 = 1/4

Eve picks the wrong basis half the time. When she does, Bob errs half the time.

~20 bits

compared publicly catch Eve about 99.7% of the time.

Limitations and what I learned

Limitations

- It is a simulation: no real photons or hardware noise
- Only one attack tested (intercept-resend)
- Real systems cannot tell noise from Eve, so they need an error threshold (about 11%)
- No error correction or privacy amplification

What I learned

- A wrong-basis measurement is what makes spying detectable
- X + H + measure is enough to encode all four BB84 states
- Fixed seeds make quantum simulations reproducible and testable

Next steps and credits

Next steps

1. Error correction (e.g. Cascade)
2. Privacy amplification
3. Test other attacks
4. Small run on real IBM hardware, reported separately

Credits

- Bennett and Brassard (1984), the BB84 protocol
- Qiskit and Qiskit Aer documentation
- Claude helped undrstand syntax while dealing with functions usined in libraries. It also helped in writing code to generate images based on the output rather than generating images using AI. It also helped in understanding and debugging errors.